

CL-TAV algorithm specification package

CL-TAV specification (CR-2026-014 / DD-036)

This package exists only to typeset the CL-TAV algorithm specification. It is not a second program and selects no solver technology. **ALG-CLTAV-01** is the top-level session; **ALG-CLTAV-02/03/04** carry the S2, S6 and S7–S8 detail; Appendix A–C carry the interface contracts, decision tables and unresolved kernels. Step labels **S0–S10** are the stable bilingual mapping; typeset line numbers are not cited.

Algorithm 1: CL-TAV top-level session (ALG-CLTAV-01)

```
Require: declared  $H_0 = \{h_{\text{normal}}\} \cup H_{\text{single}}$ ; SessionContext  $\Gamma$  (unique state); HistoryHandle  $\eta$ ; exactly one resource mode  
BUDGET( $c_{\text{min}}, B$ ) xor ROUNDS( $K_{\text{max}}$ ); action library; measurement uncertainty  
Ensure: stop class, final  $H$ ,  $\eta$ , session trace; singleton  $h_{\text{normal}}$  is not protocol PASS  
// S0 initialize the session carriers;  $q$  is  $\Gamma.\text{currentSummary}$  and  $\text{retry}$  is  $\Gamma.\text{retryCount}$ ; decision index  $k$ ,  
billed charges and  $\text{retry}$  are distinct  
1  $(\Gamma, \eta, H) \leftarrow \text{Initialize}(H_0, \text{library}, \text{uncertainty}, \text{resourcePolicy})$ ;  
2 while true do  
    // S1 entry stop gate before prediction; an already decided Admit A2--A5 or P1--P5 returns here  
3  $\text{stop} \leftarrow \text{EntryStop}(\Gamma, \eta, H)$ ;  
4 if  $\text{stop}$  is decided then return  $\text{Finish}(\text{stop}, \Gamma, \eta, H)$ ;  
    // S2 predict before select; a PredictionGapError is TEST-scoped and does not by itself block eligible  
    Recover or Prep  
5  $\text{prediction} \leftarrow \text{PredictCurrent}(\Gamma, \eta, H, \text{library}, \text{uncertainty})$ ;  
6 if  $\text{prediction}$  is PredictionGapError and Recover and Prep are ineligible then return  $\text{Finish}(\text{prediction-gap}, \Gamma, \eta, H)$ ;  
    // S3 select the final non-ADMIT action, then freeze the immutable snapshot  
7  $\text{decision} \leftarrow \text{SelectAndAdmit}(\Gamma, \eta, H, \text{library}, \text{prediction})$ ;  
8 if  $\text{decision}$  is an exit then return  $\text{Finish}(\text{decision}, \Gamma, \eta, H)$ ;  
    // FREEZEFINALSELECTION: freeze the final non-ADMIT action only after all selection rules  
9  $q_{\text{UsedAtSelect}} \leftarrow \Gamma.\text{currentSummary}$ ;  $\text{histVer} \leftarrow \eta.\text{version}$ ;  $\text{classesUsedAtSelect} \leftarrow C$ ;  $\text{actionId} \leftarrow t^*$ ;  
    // S4 charge once;  $\text{retry cap} \neq$  budget exhaustion  
10  $\Gamma \leftarrow \text{ChargeOnce}(\Gamma, \text{actionId})$ ;  
    // S5 execute the admitted action exactly once  
11  $\text{record} \leftarrow \text{ExecuteAndRecord}(\Gamma, \text{actionId})$ ;  
    // S6 interpret the execution record  
12  $\text{outcome} \leftarrow \text{InterpretOutcome}(\text{record}, \text{actionId}, \Gamma, \eta)$ ;  
    // S7--S8 classify the effect and resolve retry or confirmed recovery  
13  $(\Gamma, \eta, \text{disposition}) \leftarrow \text{ResolveOutcome}(\text{outcome}, \text{actionId}, \Gamma, \eta)$ ;  
14 if  $\text{disposition}$  is STOP then return  $\text{Finish}(\text{reason}, \Gamma, \eta, H)$ ;  
15 if  $\text{disposition}$  is RETRY then continue;  
    // S9 history reads the select snapshot; this is the sole commit of a confirmed successor summary  
16 if  $I_z$  is a valid observation class or  $\text{summaryConfirmed}$  is true then  
17      $(\eta, H) \leftarrow \text{IF-HIST-UPDATE}(\eta, H, \text{actionId}, q_{\text{UsedAtSelect}}, \text{classesUsedAtSelect}, I_z, \text{postSummary})$ ;  
18     if  $\Gamma.q\text{Status}$  is KNOWN and  $\text{summaryConfirmed}$  is true and  $\text{postSummary}$  is confirmed then  
19          $\Gamma.\text{currentSummary} \leftarrow \text{postSummary}$ ;  
    // S10 post-update exclusive P-stops; if none, the next iteration starts at S1  
20  $\text{stop} \leftarrow \text{PostUpdateStop}(\Gamma, \eta, H)$ ;  
21 if  $\text{stop}$  is decided then return  $\text{Finish}(\text{stop}, \Gamma, \eta, H)$ ;
```

Appendix A. Interface contracts

The interface registry `configs/research/cltav_interface_registry.json` is the structural authority. The table below is the typeset view of the same contracts; the procedure column names the wrapper that carries the call in this package.

Algorithm 2: PredictCurrent: current-observation prediction wrapper (ALG-CLTAV-02, S2)

Require: SessionContext Γ ; HistoryHandle η ; current H ; action library; common measurement uncertainty
Ensure: currently valid nonempty observation classes $\mathcal{C}$, or a named PredictionGapError; the session's own η is read, not rebuilt
// read the current session summary only while it is known; do not rebuild H_0
1 $classes \leftarrow \text{IF-PRED-OBS}(\Gamma, \eta, H, \text{uncertainty}, \text{library})$;
// project declared classes onto current H ; an empty class from an excluded hypothesis is not a surviving
 $score = 0$ class
2 $\mathcal{C} \leftarrow \text{ProjectOntoCurrentH}(classes, H)$;
3 **if** $\mathcal{C}$ has no nonempty member **then**
 // no currently valid class: a named prediction-gap; this is TEST-scoped and does not by itself block
 eligible Recover or Prep
4 **return** PredictionGapError;
5 **return** $\mathcal{C}$, carrying the explicit gap status rather than defaulting the backend result to the empty set;

Algorithm 3: InterpretOutcome: execution and observation interpretation (ALG-CLTAV-03, S6)

Require: execution result; immutable select snapshot ξ (see the symbol table); session and history context; declared measurement uncertainty
Ensure: structured outcome: execution effect class, observation validity and feasible classes, whether the successor summary is confirmed, the confirmed successor $postSummary$, whether a Prep/Recover confirmation actually ran, $targetConfirmed$, $prepErr$, and evidence references
// CONFIRMED-NOT-SENT and UNKNOWN-EFFECT do not call an observation or confirmation interface that was not executed; an unset confirmation field is not an evaluated failure
1 **if** $effect \in \{\text{CONFIRMED-NOT-SENT}, \text{UNKNOWN-EFFECT}\}$ **then**
2 skip interpretation; keep the successor summary unconfirmed;
3 **else**
4 $(I_z, effect, summaryConfirmed, postSummary) \leftarrow \text{IF-OBS-INTERPRET}(\Gamma, record)$;
 // a normal TEST explicitly receives summaryConfirmed and postSummary; event ownership, pairing and interval treatment follow the method contract
5 **if** $kind \in \{\text{PREP}, \text{RECOVER}\}$ and $effect \notin \{\text{CONFIRMED-NOT-SENT}, \text{UNKNOWN-EFFECT}\}$ **then**
6 $(targetConfirmed, summaryConfirmed, prepErr, declaredTarget, evidence, postSummary) \leftarrow$
 $\text{IF-PREP-RECOVER}(\Gamma, \eta, actionId, record)$; $prepResultEvaluated \leftarrow \text{true}$;
 // IF-PREP-RECOVER returns values; it does not write Γ or η . A Prep whose target is not reached may still have a confirmed actual successor
7 **if** $kind$ is Prep and $prepResultEvaluated$ and $summaryConfirmed$ is false **then** record an unconfirmed Prep successor summary for $S7$;
8 ;
9 **return** the structured outcome, including whether the unconfirmed Prep successor condition ($kind$ is Prep and $prepResultEvaluated$ and $summaryConfirmed$ is false) holds;

The short wrapper procedures below are the only places the interface macros are invoked. They add no semantics beyond DD-029 / DD-033.

Appendix B. Admit and stop tables

Appendix C. Unresolved computation kernels

Control logic, call order, state reads/writes, budget, return values and error branches are fully specified above. The internal solution techniques remain open and are abstracted by named contracts; no probability default, HMM/DTMC/Bayesian kernel or continuous-time discrete grid is introduced here.

Specification logic is reviewable from this package. A stub walk-through is reproducible from the bounded session-loop spec script. The real tool and any confirmatory experiment are not implemented. Wrapper code and finite walk-throughs are not performance evidence for the algorithm.

Algorithm 4: ResolveOutcome: outcome resolution and confirmed recovery (ALG-CLTAV-04, S7–S8)

Require: structured outcome z ; immutable select snapshot ξ ; SessionContext Γ ; HistoryHandle η
Ensure: control class COMMIT / REPLY / STOP; updated Γ, η

```

1 if  $effect \in \{CONFIRMED-NOT-SENT, UNKNOWN-EFFECT\}$  or  $prepErr$  or ( $kind$  is Prep and  $prepResultEvaluated$  and  $summaryConfirmed$  is false) or ( $actionId$  is Recover and  $prepResultEvaluated$  and  $targetConfirmed$  is false) then
    // S7 classify the effect; do not fold CONFIRMED-NOT-SENT into UNKNOWN
    do not exclude  $H$ ;
    3 if  $prepErr$  or  $effect = UNKNOWN-EFFECT$  or ( $kind$  is Prep and  $prepResultEvaluated$  and  $summaryConfirmed$  is false) then
        mark  $\eta$  conservative-unknown;  $\Gamma.qStatus \leftarrow UNKNOWN$ ; clear  $\Gamma.currentSummary$ ;
    5 else if  $effect = CONFIRMED-NOT-SENT$  then
        6 leave  $\Gamma.qStatus$  and compatible history unchanged;
    7 else
        // unconfirmed Recover target
        8  $\Gamma.qStatus$  stays UNKNOWN; clear  $\Gamma.currentSummary$ ; mark  $\eta$  conservative-unknown;
    9  $\Gamma.retryCount \leftarrow \Gamma.retryCount + 1$ ;
    10 if  $\Gamma.retryCount \geq \Gamma.retryCap$  then return STOP(Stop-Error); this is decided before A4/A5 and is not Stop-Budget;
    11 return REPLY;
12 else if  $actionId$  is Recover then
    // S8 sole writer of KNOWN( $declaredTarget, evidence$ )
    13  $\Gamma.qStatus \leftarrow KNOWN(declaredTarget, evidence)$ ;
    14  $postSummary \leftarrow (declaredTarget, evidence)$ ;
15 return COMMIT; no path here starts prediction directly, S9 remains the successor-summary commit;

```

Algorithm 5: Select and admit (contract for ALG-CLTAV-01 S3)

Require: $\Gamma; \eta$; current H ; action library; predicted classes $\mathcal{C}$
Ensure: kind; final t^* ; selectable S ; an Admit A2–A5 exit

```

1  $A \leftarrow$  currently enabled actions;
2 ( $kind, t^*, S, admit$ )  $\leftarrow$  IF-SELECT-ADMIT( $A, \Gamma.currentSummary, \mathcal{C}, \Gamma, \eta, H$ );
3 if  $kind = ADMIT$  then
    4 return the Admit A2–A5 exit;
5 else if  $kind = TEST$  then
    6 if  $S = \emptyset$  then return a named spec error; never argmin an empty  $S$ ;
    7 ;
    //  $t^*$  is the final one-step minimax action on currently valid nonempty classes inside TEST-only  $S$ 
8 else if  $kind \in \{PREP, RECOVER\}$  then
    9 keep the Prep or Recover  $t^*$  named by IF-SELECT-ADMIT; do not enter TEST minimax;
10 return  $kind, t^*$ ;

```

Chinese step groups (same process, not a second program)

- S0: Γ 为唯一状态源; q 是 $\Gamma.currentSummary$ 的别名, $qStatus$ 只记录该摘要是否已知; $retry / R_{max}$ 分别是 $retryCount / retryCap$ 的别名; 初始化 η, H_0 与 $retry$ 。
- S1: 停止 / 错误优先门; 已决定的 A2–A5 或 P1–P5 立即返回。
- S2: 选择前预测 (ALG-CLTAV-02); PredictionGapError 只关掉 TEST 评分; UNKNOWN 且 Recover 仍合格时不得挡合法恢复。
- S3: 互斥选择 TEST minimax / Prep / Recover / Admit; 空 TEST 集不得 argmin; 空 $\mathcal{C}$ 不阻止合格 Recover。选择后、任何写接口之前冻结 $qUsedAtSelect$ 、历史版本、预测类与动作身份。
- S4: 一次计费; 重试上限不是预算耗尽。决策索引 k 、计费次数与 $retry$ 彼此不同。
- S5–S6: 执行与解释返回效果类 (ALG-CLTAV-03); 正常解释还显式返回后继摘要是否确认。CONFIRMED-NOT-SENT 绕过解释与 Prep 确认, 不把它和 UNKNOWN-EFFECT 合成同一种未知。
- S7–S8: 结果处置 (ALG-CLTAV-04); $prepErr$ 、UNKNOWN-EFFECT、后继摘要未确认的 Prep, 以及目标未确认的 Recover, 均将 η 标为保守未知并清除 q ; CONFIRMED-NOT-SENT 保持 q 与相容历史并仍计重试。达 R_{max} 返回 Stop-Error。

Interface	Duty (read-only)	Outputs	Carried by
IF-PRED-OBS	finite conservative current-observation abstract	currently valid nonempty classes, or PredictionGapError	PredictCurrent (ALG-CLTAV-02)
IF-SELECT-ADMIT	exclusive TEST / Prep / Recover / Admit, final one-step minimax action	kind, selectable S , Admit A2–A5, final t^*	SelectAndAdmit (this appendix)
IF-EXECUTE-RECORD	issue the admitted action once	record, effect class, correlation id	ExecuteAndRecord (this appendix)
IF-OBS-INTERPRET	correlation, pairing, interval validity, effect class	I_z , effect class, summaryConfirmed , postSummary	InterpretOutcome (ALG-CLTAV-03)
IF-PREP-RECOVER	declare target and successor confirmation separately	targetConfirmed , summaryConfirmed , prepErr , declaredTarget , evidence , postSummary	InterpretOutcome (ALG-CLTAV-03)
IF-HIST-UPDATE	advance from the select-time snapshot; whole-history path	next η , $H' \subseteq H$ or Stop-Empty	S9 of ALG-CLTAV-01
IF-EQUIV	return established only with a valid proof	established / not-established / unknown	StopGate (this appendix)
IF-RESOURCE-STOP	A1–A5 / P1–P5, charging, return sets, residual obligations	stop class, final H , trace	StopGate (this appendix)

Algorithm 6: Execute and record (contract for ALG-CLTAV-01 S5)

```

1 (record, effect)  $\leftarrow$  IF-EXECUTE-RECORD( $\Gamma$ , actionId); only the admitted action is issued once;
2 return record, effect;

```

- S9: 历史更新读选择时快照 $qUsedAtSelect$ 与单独的已确认 $postSummary$, 不读效果后的摘要; 仅在 ‘qStatus’ 为 KNOWN 且后继摘要确认时提交 $\Gamma.currentSummary$ 。
- S10: 互斥 P1–P5; 未停止则下一轮从 S1 继续。未确认恢复回边在 S7, 不代替这条正常循环。

Stop-645 在本增量之后表示仍未在能力/实现层关闭的具名 645 依赖义务, 不是 “645 文件缺失”。

Algorithm 7: Stop gate (contract for ALG-CLTAV-01 S1 and S10)

Require: Γ ; H ; η ; named 645 residuals; equivalence status
Ensure: an exclusive stop class, or *undecided*
1 $\sigma \leftarrow$ first of Stop-Empty, Stop-Singleton, Stop-645, Stop-Equivalent, Stop-Budget via **IF-RESOURCE-STOP**(Γ, H, η) and **IF-EQUIV**(Γ, H, η);
2 **if** σ *is set* **then return** σ ;
3 **return** *undecided*; a currently non-distinguishing H is not proved equivalent;

Admit	Meaning
A1	S nonempty: execute the chosen action
A2	KNOWN: a strictly-reducing TEST or Prep exists but is unaffordable: Stop-Budget
A3	KNOWN: no strictly-reducing TEST and no Prep: Stop-NoDistinguisher
A4	UNKNOWN: Recover not eligible: Stop-Error
A5	UNKNOWN: Recover eligible but unaffordable: Stop-Budget

Symbol	Meaning
$\bar{H}$	candidate set; $H_0 = \{h_{\text{normal}}\} \cup H_{\text{single}}$
Γ	SessionContext, the unique state holder; $q = \Gamma.\text{currentSummary}$
η	HistoryHandle; $\eta.H$ and H are one logical object, not two drifting sets
ξ	immutable select snapshot: <code>actionId</code> , <code>qUsedAtSelect</code> , <code>historyVersionUsedAtSelect</code> , <code>classesUsedAtSelect</code>
z	structured outcome (effect class, I_z , <code>summaryConfirmed</code> , <code>postSummary</code> , confirmation flags, <code>prepErr</code>)
δ	ResolveOutcome control class COMMIT / RETRY / STOP

Module	Abstract kernel	Later implementation gate
PredictCurrent	finite conservative observation abstract and partition	estimator/partition algorithm
IF-HIST-UPDATE	whole-history compatible-state representation and solver	state/clock representation and solver
IF-OBS-INTERPRET	event collection and clock error budget	collectors and timestamp handling
InterpretOutcome	successor-summary confirmation	protocol confirmation contract
IF-PREP-RECOVER	device recovery	protocol/device recovery steps
IF-EQUIV	provable equivalence decision	optional proof algorithm; not required this round
IF-EXECUTE-RECORD	protocol execution adapter, time-out, I/O	protocol adapter and I/O